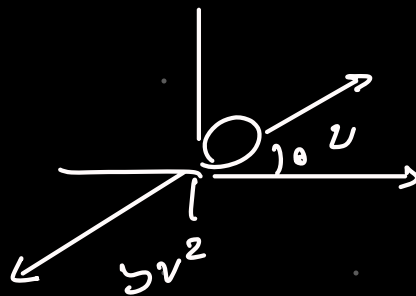


$$A = \underbrace{+bv}_{\text{viscoso}} + \underbrace{bv^2}_{\text{previsso}}$$

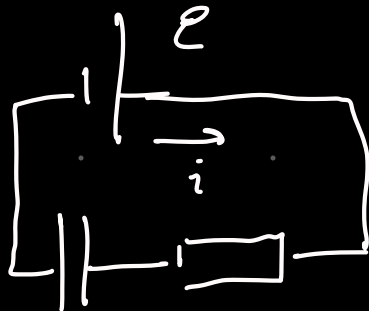


Previsso em movimento acoplado



↓
depende mas
outros eixos
nem se pode
separar os eixos

Aumento Viscoso



analogia elétrica

$$Ri + \frac{Q}{C} = E$$

$$R \frac{di}{dt} + C \frac{dQ}{dt} = \cancel{\frac{dE}{dt}}$$

$$\frac{di}{dt} = - \frac{i}{RC}$$

$$i = i_0 e^{-t/RC}$$

$$U = E(1 - e^{-t/RC})$$

$$\frac{Q}{C} = i_0 R (1 - e^{-t/RC})$$

$$Q = i_0 RC (1 - e^{-t/RC})$$

$$Q = \int i dt$$

Arredondado viscoso em x

$$x: m \cdot a_x = -b v_x$$

$$\frac{dv_x}{dt} = -\frac{b}{m} v_x$$

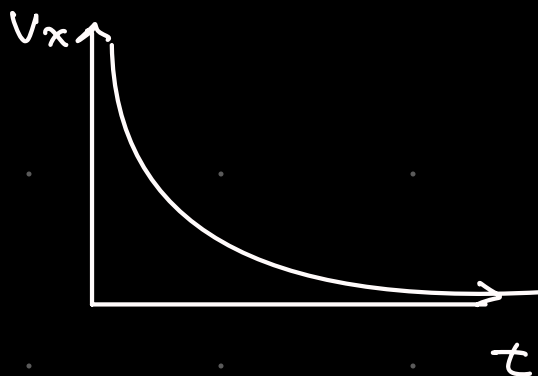
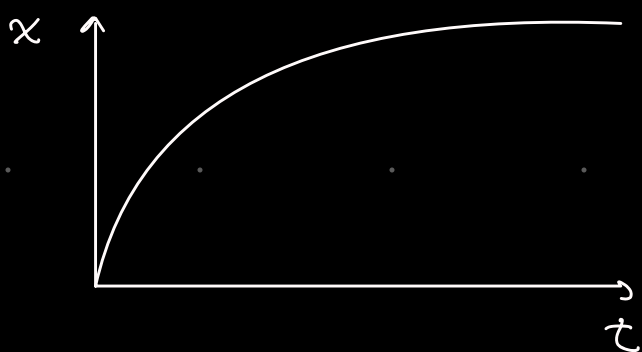
$\tau: \frac{m}{b}$ é tempo característico

$$\frac{dv_x}{dt} = -\frac{v_x}{\tau}$$

$$v_x = v_{0x} e^{-t/\tau} \quad v_x, t \rightarrow \infty = 0$$

$$dx = v_x dt \quad \Delta x = v_{0x} \tau (1 - e^{-t/\tau})$$

$$x_{t \rightarrow \infty} = v_{0x} \tau$$



$$\tau = \frac{m}{b} \rightarrow \text{inércia}$$
$$\tau = \frac{m}{b} \rightarrow \text{viscosidade}$$

arrivato Viscoso in g

Però quando $v_y < 0$

$$m \alpha_y = -mg - b v_y$$

$$\frac{dv_y}{dt} = -g - \frac{v_y}{\tau}$$

$v_{\text{terminal}} \quad \frac{dv_y}{dt} = 0$

$$v_{\text{terminal}}(y) = -g\tau$$

movimento in

$$\frac{dv_y}{dt} = -g - \frac{v_y}{\tau}$$

Mutazione di variabile

Definendo

$$U = v_y - v_{\text{terminal}}$$

$$\frac{d(U + v_{\text{ter}})}{dt} = -g - \frac{U + v_{\text{ter}}}{\tau}$$

$$U = v_y + g\tau$$

$$v_y = U - \underbrace{v_{\text{ter}}}_{\text{cte}}$$

$$\frac{d(U + v_{\text{ter}})}{dt} = -g - \frac{(U - g\tau)}{\tau}$$

$$\frac{dU}{dt} = -g + g - \frac{U}{\tau} \Rightarrow$$

$$\frac{dU}{dt} = -\frac{U}{\tau}$$

$$U = U_0 e^{-t/\tau_0} \quad U = V_y - V_{\text{ter}} \\ U_0 = V_{0y} - V_{\text{ter}}$$

$$V_y - V_{\text{ter}} = (V_{0y} - V_{\text{ter}}) e^{-t/\tau_0}$$

$$V_y = V_{\text{ter}} + (V_{0y} - V_{\text{ter}}) e^{-t/\tau_0}$$

$$V_y = V_{0y} e^{-t/\tau_0} + V_{\text{terminal}} (1 - e^{-t/\tau_0})$$

$$\Delta y = \int V_y dt = \int V_{\text{ter}} dt + \int (V_{0y} - V_{\text{ter}}) e^{-t/\tau_0} dt$$

$$\Delta y(t) = V_{\text{terminal}} t + (V_{0y} - V_{\text{ter}}) \tau_0 (1 - e^{-t/\tau_0})$$

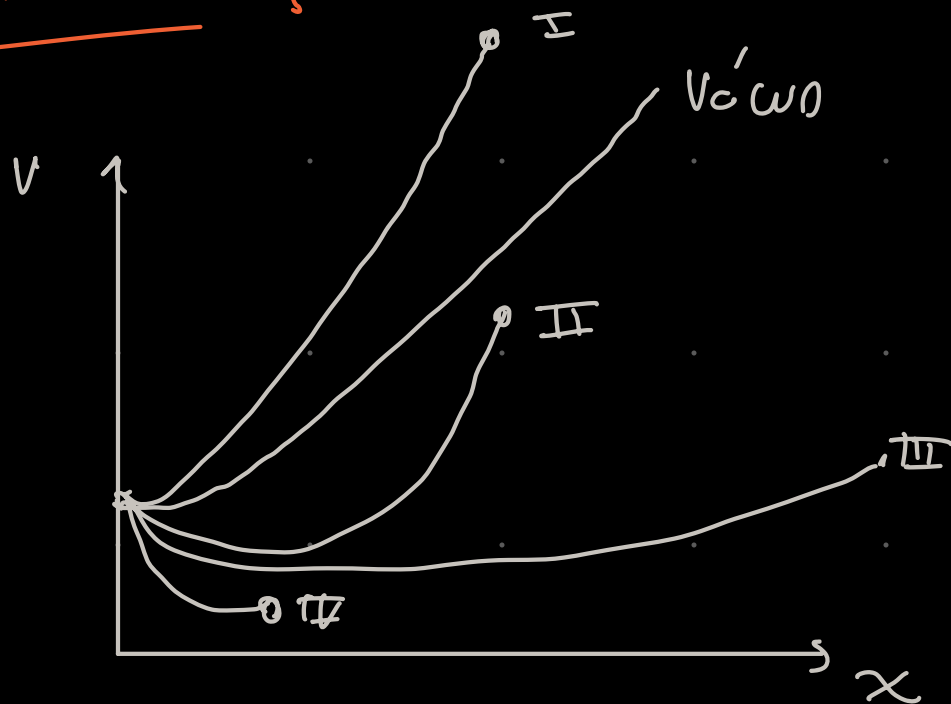
Lancamento horizontal $V_{0y} = 0$

$$V_y = V_{\text{ter}} (1 - e^{-t/\tau_0})$$

$$\Delta y(t) = V_{\text{terminal}} (t - \tau_0 (1 - e^{-t/\tau_0}))$$

$$V_y = \frac{-g}{V_{0x}} \cdot x$$

HTA 2023 Q1



$$v_y = \frac{-g}{v_0 x} \cdot x \quad \sqrt{v_y^2 + v_x^2}$$

$$v_x = v_{0x} - \frac{x}{\tau_0} \sqrt{\left(\frac{-g}{v_0 x} \cdot x\right)^2 + \left(v_0 x - \frac{x}{\tau_0}\right)^2}$$

$$|v| = \sqrt{\frac{g^2 x^2}{v_0 x^2} + \cancel{v_0 x^2} + \frac{x^2}{\tau_0^2} - \frac{2v_0 x x}{\tau_0}}$$

(Note: The terms $\cancel{v_0 x^2}$ and $\cancel{v_0 x x}$ are crossed out in the original image)

$$|v| = \sqrt{\left(\frac{g^2}{v_0 x^2} + \frac{1}{\tau_0^2}\right) x^2 - \left(\frac{2v_0 x}{\tau_0}\right) x + v_0 x^2}$$

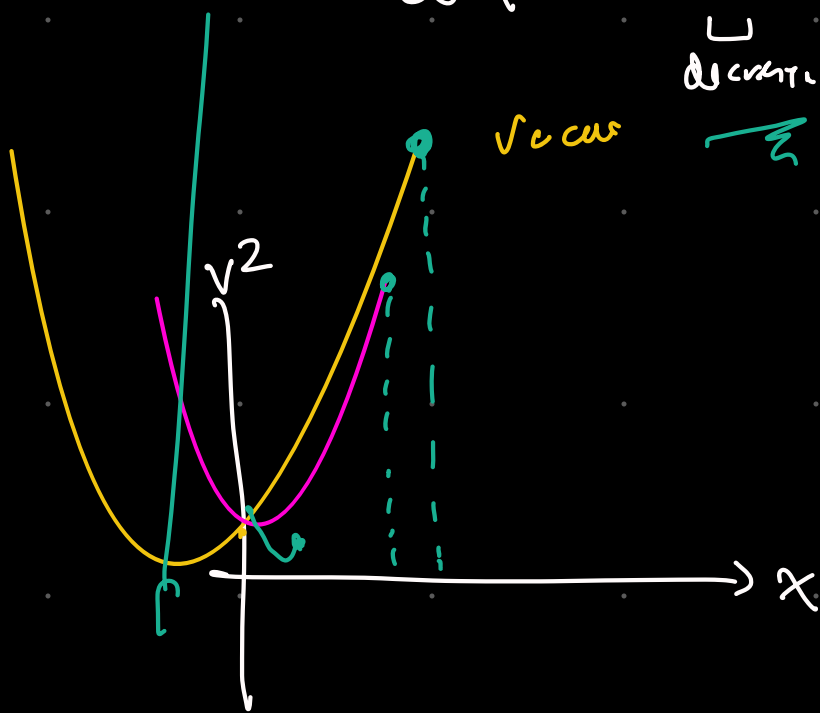
(Note: The term $\frac{g^2}{v_0 x^2}$ is underlined in the original image)

$$ax^2 - bx + c$$

discriminante

para $x=0$

é decrescente



para $v=0$

$$\frac{1}{b} = 0$$

para:

fora do decrescente

$a \uparrow$

$b \downarrow$

$\textcircled{\text{II}}$ ou II

$$h = g \tau^2 e^{-\tau/\tau_0}$$

$$-\tau_0 \ln \frac{h}{g \tau^2} = \tau$$

$$\frac{dv_y}{dt} = +g - \frac{v_y}{\tau}$$

↑ (-) ↓ (-)

$$v_{\text{terminal}} = g\tau$$

$$\lambda = v_T - v_y$$

$$v_T - \lambda = v_y$$

$$v_T = g\tau$$

$$\frac{d(v_T - \lambda)}{dt} = g - \frac{(g\tau - \lambda)}{\tau}$$

$$-\frac{d\lambda}{dt} = \frac{\lambda}{\tau}$$

Habitaculo in Coliforum.

Salvete, vivis et vivis

Salva tu

Quid agis vivis

Ita, ego in urbe habit